



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-021 Data Ingestion & Interfaces - Practice Pack

14 genuine-attempt tasks | teacher walkthrough first

Artifact	C3_021_DATA_INGESTION_INTERFACES_PRACTICE_PACK_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop. Videos are visual companions; the book remains sufficient fallback.
C2B boundary	Bridge foundations may be assumed, but C3 engineering behavior is demonstrated, broken, repaired, transferred and validated.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

Course rule: preserve what arrived -> validate interpretation -> quarantine ambiguity -> publish durably -> commit state last -> prove replay safety. A green job is never sufficient evidence by itself.



Practice protection rule

Each Pxx starts only after you followed the matching lesson once. Save `evidence/Pxx/attempt\_001/` with expected behavior, actual output/state, independent validation, failure/uncertainty and explain-back. Freeze it before opening C3-022. After review, close it and complete the changed mastery task. Never overwrite attempt 001.

Normal practice may use the lesson, official references and hint ladder. Do not use the protected review before a genuine saved attempt.



P01 - Ingestion architecture: source -> landing -> staging

Matches C3-01-L01 | Teacher walkthrough already completed at lesson pages 3-5.

YOUR INDEPENDENT TASK

Use `orders\_nested.json` instead. Define a fresh source contract and manifest without copying CSV-specific fields.

STARTING ASSET / TOOL

Practice/data/customers_messy.csv + Python csv

EXPECTED RESULT

A byte count and stable hash prefix print. Re-running without changing the file produces the same hash.

INDEPENDENT VALIDATION

Prove raw hash/path exists; accepted + quarantine = source records; staging path is different from landing; manifest has UTC run time.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P01/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R01 open.

CHANGED MASTERY TASK

After review, change source type and one contract field. Rebuild from blank without looking at the reviewed attempt.



P02 - CSV ingestion and defensive parsing

Matches C3-01-L02 | Teacher walkthrough already completed at lesson pages 6-8.

YOUR INDEPENDENT TASK

Create a fresh semicolon-delimited file with one quoted semicolon and one invalid number. Configure the parser deliberately and quarantine only the invalid row.

STARTING ASSET / TOOL

Practice/data/customers_messy.csv + Python csv

EXPECTED RESULT

The quoted comma in a name remains inside one `name` value; headers print once and rows are dictionaries rather than manually split lists.

INDEPENDENT VALIDATION

accepted + quarantine = source data rows; required headers present; keys unique where required; two manual raw-to-output comparisons; quoted delimiter preserved.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P02/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R02 open.

CHANGED MASTERY TASK

Use a different delimiter/blank pattern and rebuild without opening the review.



P03 - JSON and nested-record ingestion

Matches C3-01-L03 | Teacher walkthrough already completed at lesson pages 9-11.

YOUR INDEPENDENT TASK

Add a nested `shipping` object and preserve its fields without changing item grain; then add an order with an empty items list.

STARTING ASSET / TOOL

Practice/data/orders_nested.json + Python json

EXPECTED RESULT

Two counts print: number of parent orders and total child items. Your normalized outputs should later reconcile to exactly those grains.

INDEPENDENT VALIDATION

parent output count = source orders; child count = total source items; each child has parent key; parent total reconciles once per order.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P03/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R03 open.

CHANGED MASTERY TASK

Normalize a fresh customer->addresses document into customer and address grains.



P04 - Parquet ingestion and metadata inspection

Matches C3-01-L04 | Teacher walkthrough already completed at lesson pages 12-14.

YOUR INDEPENDENT TASK

Generate a second real Parquet file with one additional nullable column. Compare schemas and decide whether the change is compatible.

STARTING ASSET / TOOL

P04_parquet + PyArrow

EXPECTED RESULT

The version command prints a PyArrow version. A real Parquet file is created. Metadata and projected read report the same record count.

INDEPENDENT VALIDATION

PyArrow version recorded; file opens as Parquet; metadata row count = projected row count; projection contains exactly requested columns; schema difference explained.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P04/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R04 open.

CHANGED MASTERY TASK

Build a new Parquet from a changed CSV and inspect one different two-column projection.



P05 - REST APIs: requests, status codes and contracts

Matches C3-01-L05 | Teacher walkthrough already completed at lesson pages 15-17.

YOUR INDEPENDENT TASK

Add a client call that requires a new key from `/profile`; deliberately request a nonexistent key first and show your contract failure.

STARTING ASSET / TOOL

P05_P07_mock_api + Requests

EXPECTED RESULT

The health request returns JSON indicating service health. The authenticated profile succeeds; deliberately missing auth produces 401 rather than silent acceptance.

INDEPENDENT VALIDATION

timeout set; expected status checked; JSON/content checked; required keys checked; token absent from source/log; missing-auth case fails clearly.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P05/attempt_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R05 open.

CHANGED MASTERY TASK

Use another mock endpoint and define its contract from scratch.



P06 - API authentication, pagination and rate limits

Matches C3-01-L06 | Teacher walkthrough already completed at lesson pages 18-20.

YOUR INDEPENDENT TASK

Change page size or mock continuation shape and prove complete extraction again.

STARTING ASSET / TOOL

P05_P07_mock_api + Requests

EXPECTED RESULT

All mock order IDs are collected exactly once. The rate-limit endpoint shows one 429 then a successful response after respecting the server's rule.

INDEPENDENT VALIDATION

landed page count known; final unique IDs = expected full set; no repeated page/cursor cycle; 429 recovery bounded; token secret scan clean.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P06/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R06 open.

CHANGED MASTERY TASK

Use a cursor-style changed mock case and re-prove complete unique extraction.



P07 - Retries, backoff, jitter and transient failures

Matches C3-01-L07 | Teacher walkthrough already completed at lesson pages 21-23.

YOUR INDEPENDENT TASK

Create a local function that fails twice, then succeeds. Run it through your retry helper and record exact attempt count; then test a permanent `ValueError` that must not retry.

STARTING ASSET / TOOL

P05_P07_mock_api + Requests/retry helper

EXPECTED RESULT

One run shows 503 -> 503 -> success within the budget; another deliberately small budget ends with a clear non-success exit/exception.

INDEPENDENT VALIDATION

retry class explicit; max attempts explicit; delay capped; jitter present; success path and bounded-failure path both evidenced; permanent error not retried.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under ``evidence/P07/attempt_001/``. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R07 open.

CHANGED MASTERY TASK

Change retryable status and max budget, then re-run without review open.



P08 - Database extraction and safe query boundaries

Matches C3-01-L08 | Teacher walkthrough already completed at lesson pages 24-26.

YOUR INDEPENDENT TASK

Extract a different window and add a tie timestamp. Prove both tied rows appear exactly once under your chosen ordering.

STARTING ASSET / TOOL

P08_database/practice.db + sqlite3

EXPECTED RESULT

Extracted rows are stably ordered, values were bound rather than concatenated, and independent count query equals extract row count.

INDEPENDENT VALIDATION

explicit columns; bound values; boundary stated; stable ordering; source count equals output; key uniqueness checked.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P08/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R08 open.

CHANGED MASTERY TASK

Use a new window/table and rebuild query + independent control from blank.



P09 - Historical batch loading and landing metadata

Matches C3-01-L09 | Teacher walkthrough already completed at lesson pages 27-29.

YOUR INDEPENDENT TASK

Add an August batch and then modify one source row. Demonstrate your landing logic distinguishes identical rerun from changed-source rerun.

STARTING ASSET / TOOL

P09_historical + hashlib/json

EXPECTED RESULT

Three historical raw objects remain addressable with distinct manifests. An unchanged rerun is recognized rather than replacing prior evidence.

INDEPENDENT VALIDATION

every batch has unique path + manifest + hash + count + source window; changed bytes yield changed fingerprint; original remains preserved.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P09/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R09 open.

CHANGED MASTERY TASK

Land a changed-source historical batch under a new run and explain why it must not overwrite old evidence.



P10 - Incremental loads and high-water marks

Matches C3-01-L10 | Teacher walkthrough already completed at lesson pages 30-32.

YOUR INDEPENDENT TASK

Add two rows with identical updated_at and distinct IDs. Inject failure after output temp file but before state replace; rerun and prove no skip/duplicate accepted state.

STARTING ASSET / TOOL

P10_incremental + json/filesystem state

EXPECTED RESULT

Rows sharing a timestamp are not skipped. A failure before state commit leaves old state unchanged; rerun replays safely and then advances once.

INDEPENDENT VALIDATION

old/new cursor logged; every eligible tuple $>$ old and $\leq$ new; source/output counts reconcile; state unchanged on failure; rerun stable.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P10/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R10 open.

CHANGED MASTERY TASK

Use a different tie key and changed two-run dataset; prove same invariants.



P11 - CDC concepts and change-event semantics

Matches C3-01-L11 | Teacher walkthrough already completed at lesson pages 33-35.

YOUR INDEPENDENT TASK

Add a create -> update -> delete -> recreate sequence for one key and predict final state before running.

STARTING ASSET / TOOL

P11_cdc/cdc_events.json + Python

EXPECTED RESULT

The final current-state keys/values match the ordered event semantics. Raw CDC events remain preserved even though current state contains only the latest state.

INDEPENDENT VALIDATION

positions ordered/unique as required; each op handled; final state manually predicted; replay yields identical final state; raw log unchanged.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P11/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R11 open.

CHANGED MASTERY TASK

Use a changed event sequence including delete/recreate; rebuild from raw events without review.



P12 - Schema drift, compatibility, temporal/time-zone correctness and quarantine

Matches C3-01-L12 | Teacher walkthrough already completed at lesson pages 36-38.

YOUR INDEPENDENT TASK

Create day3 with one compatible nullable field and one critical type change. Show the first can proceed while the critical record/batch follows your declared stop/quarantine rule.

STARTING ASSET / TOOL

P12_drift_time + datetime/json

EXPECTED RESULT

The aware +05:30 timestamp prints an equivalent UTC instant. A naive timestamp is not silently treated as local or UTC unless the contract declares that rule.

INDEPENDENT VALIDATION

schema diff recorded; policy decision explicit; UTC conversion proved; ambiguous time retained in quarantine; critical unresolved drift prevents unsafe state advance.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P12/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R12 open.

CHANGED MASTERY TASK

Use a changed schema/time case; classify it without copying prior decisions.



P13 - Checkpoint/state management and replay safety

Matches C3-01-L13 | Teacher walkthrough already completed at lesson pages 39-41.

YOUR INDEPENDENT TASK

Use P13 changed batch. Fail once after staging but before state commit. Compare hashes/keys after rerun and prove the committed result is stable.

STARTING ASSET / TOOL

P13_replay + pathlib/os.replace

EXPECTED RESULT

Injected failure leaves old checkpoint intact. Rerun may repeat work, but accepted/current output does not gain unintended duplicates, then state advances once.

INDEPENDENT VALIDATION

state_before captured; state unchanged on injected failure; rerun accepted keys stable; temp state not left as active checkpoint; final state equals last durable accepted boundary.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P13/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R13 open.

CHANGED MASTERY TASK

Move the injected failure to a different pre-commit point and re-prove safety.



P14 - Ingestion testing, reconciliation and backfills

Matches C3-01-L14 | Teacher walkthrough already completed at lesson pages 42-44.

YOUR INDEPENDENT TASK

Backfill a different historical date window and add one malformed record. Prove bounded eligibility, quarantine accounting and unchanged live state.

STARTING ASSET / TOOL

P14_backfill + pytest/hashlib

EXPECTED RESULT

Tests pass only after intended behavior is implemented. Backfill processes its bounded history while the live-state hash remains unchanged.

INDEPENDENT VALIDATION

pytest includes failure/edge cases; source accounting reconciles; keys/temporal bounds checked; live state hash unchanged; backfill state clearly separate.

BREAK / FIX EVIDENCE

Deliberately exercise at least one failure or ambiguity described in the lesson. Save what failed, why it failed, and the smallest repair; do not delete the failed evidence.

FREEZE BEFORE REVIEW

Save under `evidence/P14/attempt\_001/`. Include command/code, actual output, validation control, failure/recovery and 60-90 second explain-back note. Then mark attempt frozen; only now may C3-022 R14 open.

CHANGED MASTERY TASK

Run a changed historical window with one deliberate invalid record and re-prove all invariants.